

Product Review and Rating Management System

Titan Eye – Eyewear E-Commerce Database

1. Create Review and Rating Tables

1.1 Review

Column	Type	Constraint
Review_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Review_Text	VARCHAR2(500)	—
Review_Date	DATE	—

1.2 Rating

Column	Type	Constraint
Rating_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Rating	NUMBER	—
Rating_Date	DATE	—

2. Review Table

```

SQL> CREATE TABLE Review (
  2     Review_ID NUMBER PRIMARY KEY,
  3     Customer_ID NUMBER,
  4     Product_ID NUMBER,
  5     Review_Text VARCHAR2(500) NOT NULL,
  6     Review_Date DATE NOT NULL,
  7     CONSTRAINT fk_review_customer
  8         FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
  9     CONSTRAINT fk_review_product
 10         FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
 11 );

Table created.

```

3. Rating Table

```

SQL> CREATE TABLE Rating (
  2     Rating_ID NUMBER PRIMARY KEY,
  3     Customer_ID NUMBER,
  4     Product_ID NUMBER,
  5     Rating NUMBER NOT NULL,
  6     Rating_Date DATE NOT NULL,
  7     CONSTRAINT fk_rating_customer
  8         FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
  9     CONSTRAINT fk_rating_product
 10         FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
 11 );

Table created.

```

4.Review Data Insertion

```

SQL> INSERT INTO Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (1, 101, 101, 'The frame is stylish and comfortable.', DATE '2026-09-15');

1 row created.

SQL>
SQL> INSERT INTO Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (2, 102, 102, 'Good quality metal frame with a premium look.', DATE '2026-09-16');

1 row created.

SQL> INSERT INTO Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (3, 103, 103, 'The aviator sunglasses look attractive.', DATE '2026-09-17');

1 row created.

SQL> INSERT INTO Review
  2 (Review_ID, Customer_ID, Product_ID, Review_Text, Review_Date)
  3 VALUES
  4 (4, 104, 104, 'Nice round frame and very comfortable to wear.', DATE '2026-09-18');

1 row created.

SQL> COMMIT;

Commit complete.

```

4.1 Display Review Records

```
SQL> SELECT * FROM Review;

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
          1          101          101
The frame is stylish and comfortable.
15-SEP-26

          2          102          102
Good quality metal frame with a premium look.
16-SEP-26

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
          3          103          103
The aviator sunglasses look attractive.
17-SEP-26

          4          104          104
Nice round frame and very comfortable to wear.

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
18-SEP-26
```

5. Rating Data Insertion

```
SQL> INSERT INTO Rating
2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
3 VALUES
4 (1, 101, 101, 5, DATE '2026-09-15');

1 row created.

SQL> INSERT INTO Rating
2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
3 VALUES
4 (2, 102, 102, 4, DATE '2026-09-16');

1 row created.

SQL> INSERT INTO Rating
2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
3 VALUES
4 (3, 103, 103, 5, DATE '2026-09-17');

1 row created.

SQL> INSERT INTO Rating
2 (Rating_ID, Customer_ID, Product_ID, Rating, Rating_Date)
3 VALUES
4 (4, 104, 104, 4, DATE '2026-09-18');

1 row created.

SQL> COMMIT;

Commit complete.
```

5.1 Display Rating Records

```
SQL> SELECT * FROM Rating;
```

RATING_ID	CUSTOMER_ID	PRODUCT_ID	RATING	RATING_DA
1	101	101	5	15-SEP-26
2	102	102	4	16-SEP-26
3	103	103	5	17-SEP-26
4	104	104	4	18-SEP-26

6.Retrieve Product Review Details

```
SQL> SELECT
2     r.Review_ID,
3     r.Customer_ID,
4     r.Product_ID,
5     p.Product_Name,
6     r.Review_Text,
7     r.Review_Date
8 FROM Review r
9 JOIN Product p
10    ON r.Product_ID = p.Product_ID;
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
PRODUCT_NAME
```

```
REVIEW_TEXT
```

```
REVIEW_DA
```

```
1         101         101
Titan Classic Frame
The frame is stylish and comfortable.
15-SEP-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
PRODUCT_NAME
```

```
REVIEW_TEXT
```

```
REVIEW_DA
```

```
2         102         102
Titan Metal Frame
Good quality metal frame with a premium look.
16-SEP-26
```

```

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      3      103      103
Titan Aviator Sunglasses
The aviator sunglasses look attractive.
17-SEP-26

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      4      104      104
Titan Round Sunglasses
Nice round frame and very comfortable to wear.
18-SEP-26

```

7. Calculate Average Product Rating

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(r.Rating) AS Average_Rating
  5 FROM Product p
  6 JOIN Rating r
  7   ON p.Product_ID = r.Product_ID
  8 GROUP BY
  9     p.Product_ID,
 10     p.Product_Name;

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      101
Titan Classic Frame
      5

      102
Titan Metal Frame
      4

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      103
Titan Aviator Sunglasses
      5

```

```

      104
Titan Round Sunglasses

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      4

```

8. Identify Highly Rated Products

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(r.Rating) AS Average_Rating
  5 FROM Product p
  6 JOIN Rating r
  7     ON p.Product_ID = r.Product_ID
  8 GROUP BY
  9     p.Product_ID,
 10     p.Product_Name
 11 HAVING AVG(r.Rating) >= 4;

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      101
Titan Classic Frame
      5

      102
Titan Metal Frame
      4

```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      103
Titan Aviator Sunglasses
      5

      104
Titan Round Sunglasses
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      4
```